

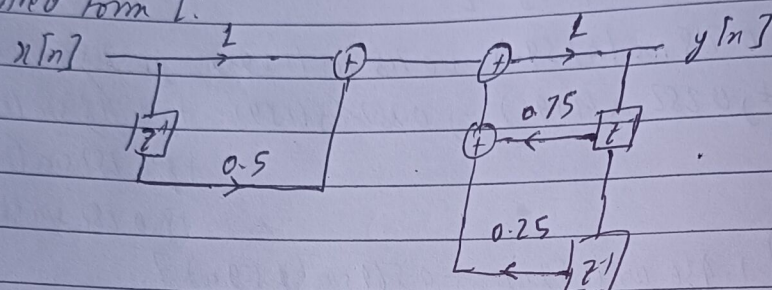
82 B

$$y[n] = 0.75 y[n-1] - 0.25 y[n-2] + x[n] + 0.5 x[n-1]$$

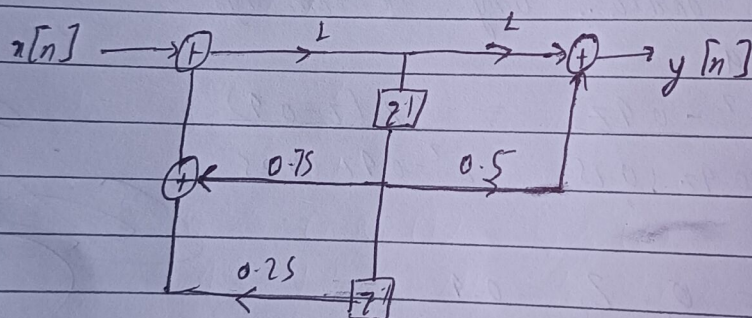
$$\text{or } Y(z) - 0.75 z^{-1} Y(z) - 0.25 z^{-2} Y(z) = X(z) + 0.5 z^{-1} X(z)$$

$$\text{or } H(z) = \frac{Y(z)}{X(z)} = \frac{1 + 0.5 z^{-1}}{1 - 0.75 z^{-1} - 0.25 z^{-2}}$$

Direct Form I:



Direct Form II:



correct arrow direction btw.

Lattice ladder:

$$H(z) = \frac{1 - 0.7 z^{-1} + 0.5 z^{-2}}{1 - 0.3 z^{-1} + 0.25 z^{-2}}$$

$$\text{So, } A(z) = 1 - 0.3 z^{-1} + 0.25 z^{-2}$$

$$B_m(z) = z^{-m} A_m(z^{-1})$$

$$= z^{-2} - 0.3 z^{-1} + 0.25$$

$B_m =$

$$B_{m-1}(z) = \frac{A_m(z) - k_m B_m(z)}{1 - k_m z^{-1}}$$

$$\alpha_2(0) = 1, \quad \alpha_2(1) = -0.3, \quad \alpha_2(2) = 0.25 = k_2$$

so,

 A_{m-1}

$$\begin{aligned} A_1(z) &= \frac{A_2(z) - k_2 B_2(z)}{1 - k_2 z^{-1}} \\ &= \frac{1 - 0.3z^{-1} + 0.25z^{-2} - 0.25(z^{-2} - 0.3z^{-1} + 0.25)}{1 - 0.25z^{-1}} \\ &= \frac{-0.275z^{-1} + 0.9375}{1 - 0.25z^{-1}} \\ &= -0.24z^{-1} + 1 \end{aligned}$$

$$\alpha_1(0) = 1, \quad \alpha_1(1) = -0.24 = k_1$$

for coefficients of all zero system:

$$\begin{aligned} B(z) &= 2 - 0.7z^{-1} + 0.5z^{-2} \\ &= 2(1 - 0.35z^{-1} + 0.25z^{-2}) \\ &= 2(1 - 0.35z^{-1} + 0.25z^{-2}) \end{aligned}$$

$$b_0 = 1, \quad b_1 = -0.35, \quad b_2 = 0.25$$

$$\text{so, } b_2 = 1_2 = 1$$

for coefficients of ladder: ~~for~~

$$Q(z) = 2 - 0.7z^{-1} + 0.5z^{-2}$$

$$= \frac{1}{2} (4 - 1.4z^{-1} + z^{-2})$$

so that:

$$Q_2 = Q_1 = 1, \quad Q_2 = 1.4, \quad Q_0 = 4$$

Now,

$$Q_m = Q_m - \sum_{i=m+1}^M c_i d_i (i-m)$$

$$\therefore Q_1 = Q_1 - \sum_{i=2}^2 c_i d_i (i-1)$$

$$= Q_1 - c_2 d_2 (1)$$

$$= 1.4 - 1 \times (-0.3)$$

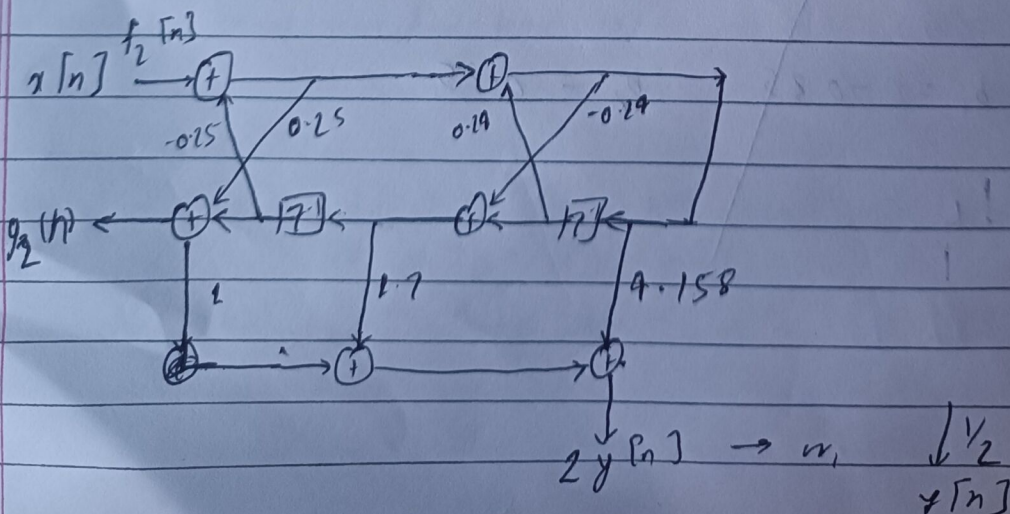
$$= 1.7$$

$$\therefore Q_0 = Q_0 - \sum_{i=1}^2 c_i d_i (i-0)$$

$$= Q_0 - c_1 d_1 (1) - c_2 d_2 (2)$$

$$= 4 - 1.7 \times (-0.24) - 1 \times (0.25)$$

$$= 4.158$$



82 Ra:

$$H(z) = \frac{0.2759 + 0.5121z^{-1} + 0.5121z^{-2} + 0.2759z^{-3}}{1 - 0.0010z^{-1} + 0.654z^{-2} - 0.0775z^{-3}}$$

$$A_3(z) = 1 - 0.0010z^{-1} + 0.654z^{-2} - 0.0775z^{-3}$$

$$\alpha_3(0) = 1, \quad \alpha_3(1) = -0.0010, \quad \alpha_3(2) = 0.654, \quad \alpha_3(3) = -0.0775 = k_3$$

$$B_3(z) = -0.0775 + 0.654z^{-1} - 0.0010z^{-2} + z^{-3}$$

$$A_2(z) = \frac{A_3(z) - k_3 B_3(z)}{1 - k_3 z^{-3}}$$

$$= \frac{1 - 0.0010z^{-1} + 0.654z^{-2} - 0.0775z^{-3} + 0.0775(-0.0775 + 0.654z^{-1} - 0.0010z^{-2} + z^{-3})}{1 + 0.00775z^3}$$

$$= \frac{0.994 + 0.049z^{-1} + 0.6539z^{-2}}{1 + 0.00775z^3}$$

$$= 1 + 0.0499z^{-1} + 0.6578z^{-2}$$

$$\alpha_2(0) = 1, \quad \alpha_2(1) = 0.0499, \quad \alpha_2(2) = 0.6578 = k_2$$

$$A_1(z) = \frac{A_2(z) - k_2 B_2(z)}{1 - k_2 z^{-2}}$$

$$= \frac{1 + 0.0499z^{-1} + 0.6578z^{-2} - 0.6578(0.6578 + 0.0499z^{-1} + z^{-2})}{1 - 0.6578z^2}$$

$$= \frac{0.5673 + 0.0171z^{-1}}{0.6578 - 0.5673}$$

$$= 1 + 0.0301z^{-1}$$

$$\alpha_1(0) = 1, \quad \alpha_1(1) = +0.0301 = k_1$$

for 1 odder core filter

$$Q_3(z) = 0.2759 + 0.5121z^{-1} + 0.5121z^{-2} + 0.2759z^{-3}$$

$$= 0.2759 [1 + 1.8561z^{-1} + 1.8561z^{-2} + z^{-3}]$$

so, $Q_0 = 1$, $Q_1 = 1.8561$, $Q_2 = 1.8561$, $Q_3 = 1$

Now,

$$C_3 = Q_3 = 1$$

$$C_2 = Q_2 - \sum_{i=m+1}^M C_i d_i (i-m) = Q_2 - \sum_{i=3}^3 C_i d_i (i-2)$$

$$= 1.8561 - C_3 d_3 (1)$$

$$= 1.8571$$

$$C_1 = Q_1 - \sum_{i=2}^3 C_i d_i (i-1)$$

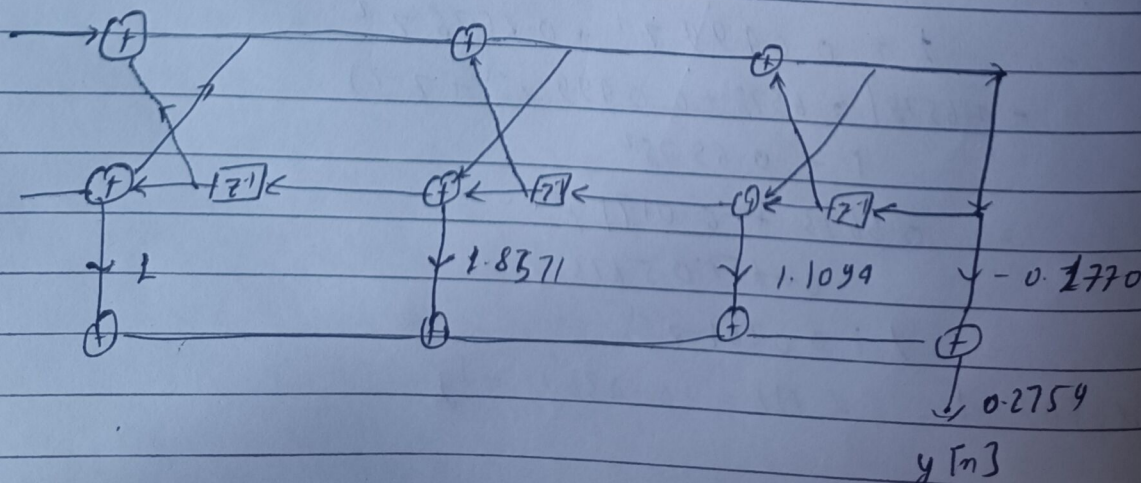
$$= \cancel{1.8561} - C_2 d_2 (1) - C_3 d_3 (2)$$

$$= 1.1094$$

$$C_0 = Q_0 - \sum_{i=1}^3 C_i d_i (i-0)$$

$$= 1 - C_1 d_1 (1) - C_2 d_2 (2) - C_3 d_3 (3)$$

$$= -0.1770$$



81 BH

$$H(z) = \frac{1 + z^{-1} + z^{-2}}{(1 + 0.5z^{-1})(1 + 0.3z^{-1})(1 + 0.4z^{-1})}$$

Now,

$$H(z) = \frac{1 + z^{-1} + z^{-2}}{1 + (0.5 + 0.3 + 0.4)z^{-1} + (0.15 + 0.2 + 0.12)z^{-2} + 0.06z^{-3}}$$

$$= \frac{1 + z^{-1} + z^{-2}}{1 + 1.2z^{-1} + 0.47z^{-2} + 0.06z^{-3}}$$

$$A_3(z) = 1 + 1.2z^{-1} + 0.47z^{-2} + 0.06z^{-3}$$

$$d_3(0) = 1, \quad d_3(1) = 1.2, \quad d_3(2) = 0.47, \quad d_3(3) = 0.06$$

$= k_3$

$$A_2(z) = A_3 - k_3 B_3$$

$$= \frac{1 + 1.2z^{-1} + 0.47z^{-2} + 0.06z^{-3}}{1 - k_3^2}$$

$$= \frac{1 + 1.2z^{-1} + 0.47z^{-2} + 0.06z^{-3} - 0.06(0.06 + 0.47z^{-1} + 1.2z^{-2} + z^{-3})}{1 - 0.06^2}$$

$$= \frac{0.9964 + 1.1718z^{-1} + 0.398z^{-2}}{0.9916}$$

$$= 1 + 1.1760z^{-1} + 0.3994z^{-2}$$

$$d_2(0) = 1, \quad d_2(1) = 1.1760, \quad d_2(2) = 0.3994$$

$$A_1(z) = A_2(z) - k_2 B_2(z)$$

$$= \frac{1 + 1.1760z^{-1} + 0.3994z^{-2}}{1 - k_2^2}$$

$$= \frac{1 + 1.1760z^{-1} + 0.3994z^{-2} - 0.3994(0.3994 + 1.1760z^{-1} + z^{-2})}{1 - 0.3994^2}$$

$$= \frac{0.8405 + 0.7063z^{-1}}{0.8405} = 1 + 0.8394z^{-1}$$

$$\alpha_2(0) = 1, \alpha_1(1) = 0.8344$$

$$Q_3(z) = 1 + z^{-1} + z^{-2} + 0.8344z^{-3}$$

↓

b_2

$$w, b_0 = 1, b_1 = 1, b_2 = 1$$

$$c_2 = b_2 = 1$$

$$c_1 = b_1 - \sum_{i=2}^2 c_i d_i(i-1)$$

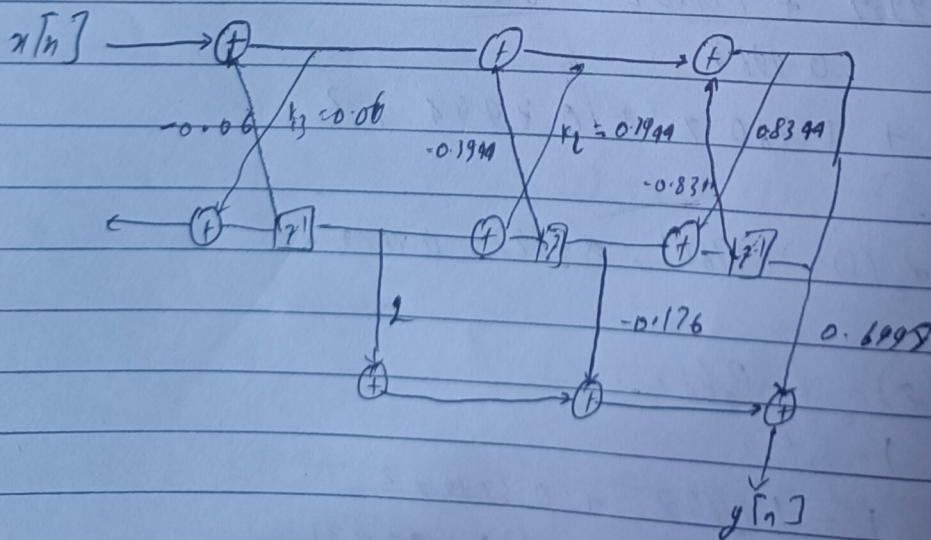
$$= 1 - c_2 d_2(1)$$

$$= -0.176$$

$$c_0 = b_0 - \sum_{i=1}^2 c_i d_i(i)$$

$$= 1 - c_1 d_1(0) - c_2 d_2(2)$$

$$= 0.6998$$



81B4

Date _____

Page _____

cascade:

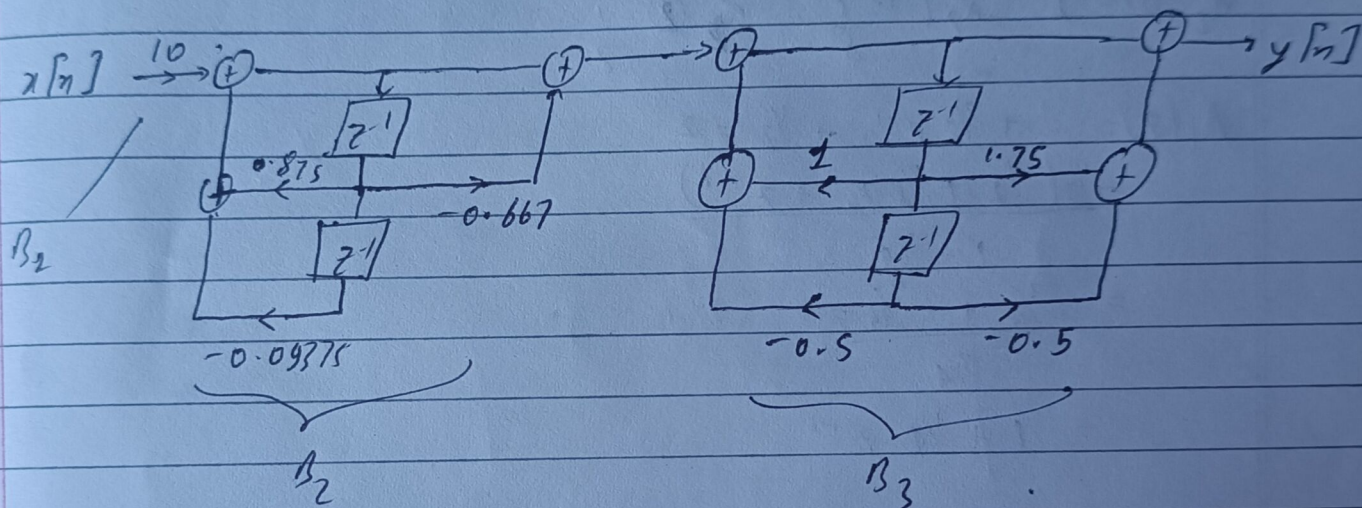
$$H(z) = \frac{10 (1 - 0.25 z^{-1}) (1 - 0.667 z^{-1}) (1 + 2z^{-1})}{(1 - 0.75 z^{-1}) (1 - 0.125 z^{-1}) (1 - (0.5 + j0.5) z^{-1}) (1 - (0.5 - j0.5) z^{-1})}$$

$$= 10 \times \frac{1 - 0.667 z^{-1}}{1 - 0.875 z^{-1} + 0.09375 z^{-2}} \times \frac{1 + 1.75 z^{-1} - 0.5 z^{-2}}{1 - z^{-1} + 0.5 z^{-2}}$$

↓
 B_1

↓
 B_2

↓
 B_3



stability

$$H(z) = 2 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3}$$

$$A_3(z) = 2 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3}$$

$$a_3(0) = 2, \quad a_3(1) = \frac{13}{24}, \quad a_3(2) = \frac{5}{8}, \quad a_3(3) = \frac{1}{3}$$

$$A_2(z) = \frac{A_3(z) - km B_3(z)}{1 - km^2}$$

$$A_2(z) = \frac{1 + \frac{13}{29}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{2}z^{-3}}{1 - \frac{1}{3}\left(\frac{1}{3} + \frac{5}{8}z^{-1} + \frac{13}{29}z^{-2} + \frac{1}{2}z^{-3}\right)}$$

$$= \frac{1 - \left(\frac{1}{3}\right)^2}{\frac{8}{9} + \frac{1}{3}z^{-1} + \frac{9}{9}z^{-2}}$$

$$= \frac{1 + \frac{3}{8}z^{-1} + \frac{1}{2}z^{-2}}{\frac{8}{9}}$$

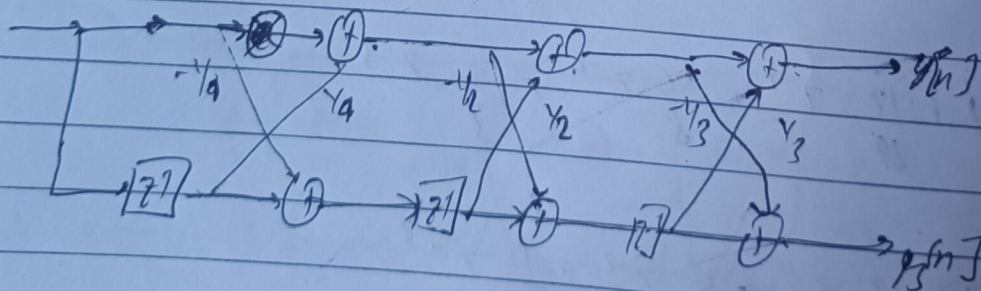
$$\alpha_2(0) = 1, \alpha_2(1) = \frac{3}{8}, \alpha_2(2) = \frac{1}{2}$$

$$A_1(z) = \frac{1 + \frac{3}{8}z^{-1} + \frac{1}{2}z^{-2}}{1 - \frac{1}{4}\left(\frac{1}{2} + \frac{3}{8}z^{-1} + z^{-2}\right)}$$

$$= \frac{\frac{3}{4} + \frac{3}{16}z^{-1}}{\frac{3}{4}}$$

$$= 1 + \frac{1}{4}z^{-1}$$

$$\alpha_1(0) = 1, \alpha_1(1) = \frac{1}{4}$$



74 Ma

Date _____

Page _____

$$H(z) =$$

$$\frac{(z - 0.75z^{-1})(z - 0.125z^{-1})}{1 - 0.875z^{-1} + 0.09375z^{-2}} \times \frac{1 + 1.75z^{-1} - 0.5z^{-2}}{1 - z^{-1} + 0.5z^{-2}}$$

parallel:

$$\begin{aligned} H(z) &= \frac{5}{z} \times \frac{z - 0.667}{(z - 0.75)(z - 0.125)} \times \frac{(z - 0.25)(z + 2)}{[z - (0.5 + j0.5)]} \\ &= 5 \times \frac{z(z - 0.667)}{(z - 0.75)(z - 0.125)} \times \frac{(z - 0.25)(z + 2)}{[z - (0.5 + j0.5)]} \end{aligned}$$

$$H(z) = \frac{A}{z - 0.75} + \frac{B}{z - 0.125} + \frac{C}{z - (0.5 + j0.5)} + \frac{C^*}{z - (0.5 - j0.5)}$$

$$\begin{aligned} A &= \frac{(z - 0.667)(z - 0.25)(z + 2)}{(z - 0.125)[z - (0.5 + j0.5)]} \Big|_{z=0.75} \\ &= 0.584 \end{aligned}$$

$$\begin{aligned} B &= \frac{(z - 0.667)(z - 0.25)(z + 2)}{(z - 0.75)[z - (0.5 + j0.5)]} \Big|_{z=0.125} \\ &= -0.589 \end{aligned}$$

$$\begin{aligned} C &= \frac{(z - 0.667)(z - 0.25)(z + 2)}{(z - 0.125)(z - 0.75)[z - (0.5 - j0.5)]} \Big|_{z=0.5 + j0.5} \\ &= 0.502 - j 2.04 \end{aligned}$$

$$C^* = 0.502 + j 2.04$$

$$H(z) = \frac{0.589}{5z} - \frac{0.589}{z-0.75} + \frac{0.502-j2.09}{z-10.5+j0.5} + \frac{0.502+j2.09}{z-10.5-j0.5}$$

$$H(z) = \frac{2.92}{1-0.75z^{-1}} - \frac{2.945}{1-0.125z^{-1}} + \frac{0.502-j2.09}{1-(10.5+j0.5)z^{-1}} + \frac{0.502+j2.09}{1-(10.5-j0.5)z^{-1}}$$

